usb_power

File: usb_power.kicad_sch

USB−C sink (no host), VBUS protection, USB0 device/FEL

power

File: power.kicad_sch

Regulators, rail order, supervisors, staged−power links

f133

File: f133.kicad_sch

F133−B SoC: supplies, clock, reset, pin assignment

boot_nor

File: boot_nor.kicad_sch

32 MiB SPI NOR boot (BOOT−SEL 00), straps, FEL/clip recovery

mn820

File: mn820.kicad_sch

MN820 (BT820B + 128 MiB DDR3L) module socket, host SPI1, LVDS to R09 panel

t35

File: t35.kicad_sch

T35F400C3 (optional accelerator): rails, config flash, SPI0 isolation, mailbox

psram

File: psram.kicad_sch

APS6404L 8 MiB PSRAM: switched VDD, switched start−up pull−downs

sd_audio

File: sd_audio.kicad_sch

microSD game media (not boot), 2x MAX98357A I2S speakers

controls

File: controls.kicad_sch

Game buttons, LEDs, debug UART

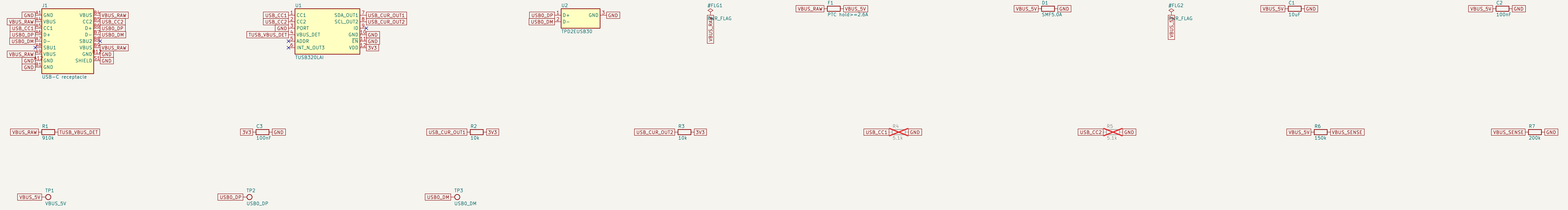
NOT approved for fabrication. LCD/panel/backlight deferred to R09.

Sheet: /
File: ARC1_R08.kicad_sch

Title: *ARC−1 Compact R08 − hierarchical root*

Size: A3
Date: 2026−10−15
Rev: R08−draft
KiCad E.D.A. 9.0.9
Id: 1/10

USB-C sink (no host), VBUS protection, USB0 device/FEL
Sheet generated from design.py; global labels carry every net.



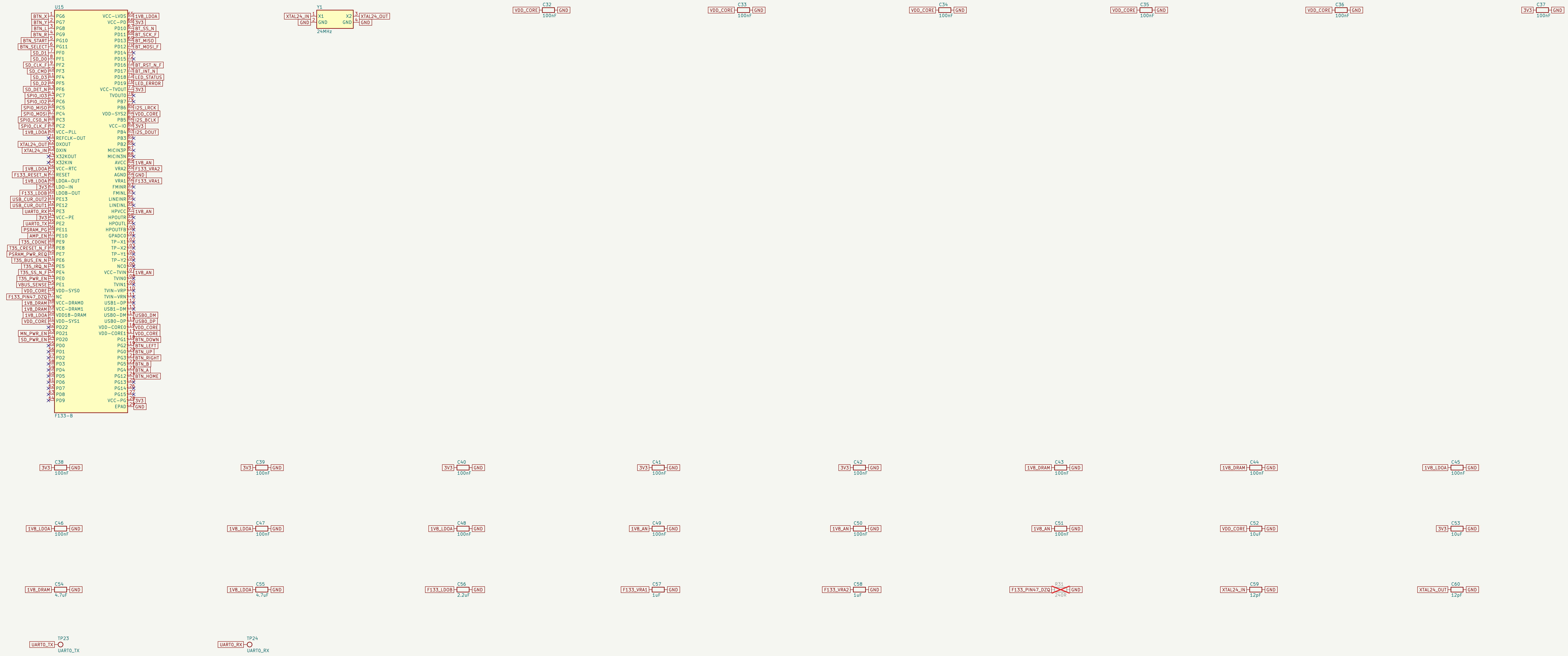
Generated by R08_WORK/cad/gen/gen_kicad.py from design.py — edit the source, not this file
SELECTED_UNVERIFIED engineering capture — NOT released for fabrication
STRUTHIO ARC−1
Sheet: /usb_power/
File: usb_power.kicad_sch
Title: ARC−1 Compact R08: USB−C sink (no host), VBUS protection, USB0 de
Size: User
Date: 2026−10−15
Rev: R08−draft
KiCad E.D.A. 9.0.9
Id: 2/10

Regulators, rail order, supervisors, staged—power links
Sheet generated from design.py; global labels carry every net.

U3 VBUS_5V 1 EN GND 2 GND SW_3V3 3 SW VIN 4 VBUS_5V PG 5 PG_3V3_BUCK FB 6 FB_3V3 3V3 2A
U4 PG_3V3 1 RESET GND 2 GND SUP33_MR_N 3 MR CT 4 SUP33_CT SENSE 5 3V3 VDD 6 3V3 TPS3808G33
U5 PG_3V3 1 EN GND 2 GND SW_CORE 3 SW VIN 4 VBUS_5V FB 5 FB_CORE VDD_CORE 0.95V
U6 PG_3V3 1 EN GND 2 GND SW_DRAM 3 SW VIN 4 VBUS_5V FB 5 FB_DRAM 1V8_DRAM
U7 3V3 1 IN GND 2 GND 3V3 3 EN NC 4 OUT 5 1V8_AN 1V8_AN
U8 F133_RESET_N 1 RESET GND 2 GND RST_BTN_N 3 MR CT 4 SUPC_CT SENSE 5 VDD_CORE VDD 6 3V3 TPS3808G09
U9 T35_PWR_EN 1 EN GND 2 GND SW_1V2 3 SW VIN 4 VBUS_5V PG 5 T35_1V2_PG FB 6 FB_1V2 1V2_T35
U10 3V3 1 VIN GND 2 GND T35_1V2_PG 3 ON CT 4 SW_T35_CT QOD 5 3V3_T35 VOUT 6 3V3_T35 3V3_T35 switch
U11 3V3 1 VIN GND 2 GND MN_PWR_EN 3 ON CT 4 SW_MN_CT QOD 5 3V3_MN VOUT 6 3V3_MN 3V3_MN switch
U12 3V3_T35 1 VIN GND 2 GND PSRAM_EN 3 ON CT 4 SW_PS_CT QOD 5 PSRAM_VDD VOUT 6 PSRAM_VDD PSRAM_VDD switch
U13 PSRAM_PWR_REQ 1 A CDONE_DLY 2 B GND 3 GND Y 4 PSRAM_EN VCC 5 3V3 AND
U14 PSRAM_PG 1 RESET GND 2 GND PSRAM_SUP_MR_N 3 MR CT 4 PSRAM_SUP_CT SENSE 5 PSRAM_VDD VDD 6 3V3_T35 TPS3808G30
SW1 RST_BTN_N 1 1 2 2 GND RESET
C4 VBUS_5V GND 10uF
L1 SW_3V3 3V3_REG 2.2uH
R8 3V3_REG FB_3V3 453k
R9 FB_3V3 GND 100k
C5 3V3_REG GND 22uF
C6 3V3_REG GND 22uF
R10 PG_3V3_BUCK 3V3_REG 100k
#FLG3 3V3_REG PWR_FLAG
R11 3V3_REG 3V3 0R
#FLG4 3V3 PWR_FLAG
TP4 3V3_REG 3V3_REG
TP5 3V3 3V3
TP6 PG_3V3_BUCK PG_3V3_BUCK
C7 SUP33_CT GND 2.2nF
R12 SUP33_MR_N 3V3 100k
R13 PG_3V3 3V3 10k
C8 3V3 GND 100nF
C9 VBUS_5V GND 10uF
L2 SW_CORE VDD_CORE_REG 2.2uH
R14 VDD_CORE_REG FB_CORE 57.6k
R15 FB_CORE GND 100k
C10 VDD_CORE_REG GND 22uF
C11 VDD_CORE_REG GND 22uF
#FLG5 VDD_CORE_REG PWR_FLAG
R16 VDD_CORE_REG VDD_CORE 0R
#FLG6 VDD_CORE PWR_FLAG
TP7 VDD_CORE VDD_CORE
C12 VBUS_5V GND 10uF
L3 SW_DRAM 1V8_DRAM 2.2uH
R17 1V8_DRAM FB_DRAM 200k
R18 FB_DRAM GND 100k
C13 1V8_DRAM GND 22uF
#FLG7 1V8_DRAM PWR_FLAG
TP8 1V8_DRAM 1V8_DRAM
C14 3V3 GND 1uF
C15 1V8_AN GND 1uF
TP9 1V8_AN 1V8_AN
C16 SUPC_CT GND 15nF
R19 RST_BTN_N 3V3 100k
R20 F133_RESET_N 1V8_LDOA 10k
TP10 F133_RESET_N F133_RESET_N
TP11 PG_3V3 PG_3V3
C17 VBUS_5V GND 10uF
L4 SW_1V2 1V2_T35_REG 2.2uH
R21 1V2_T35_REG FB_1V2 100k
R22 FB_1V2 GND 100k
C18 1V2_T35_REG GND 22uF
C19 1V2_T35_REG GND 22uF
#FLG8 1V2_T35_REG PWR_FLAG
R23 1V2_T35_REG 1V2_T35 0R
#FLG9 1V2_T35 PWR_FLAG
R24 T35_PWR_EN GND 100k
R25 T35_1V2_PG 3V3 100k
TP12 1V2_T35 1V2_T35
TP13 T35_1V2_PG T35_1V2_PG
C20 SW_T35_CT GND 1nF
C21 3V3 GND 1uF
C22 3V3_T35 GND 10uF
TP14 3V3_T35 3V3_T35
C23 SW_MN_CT GND 1nF
C24 3V3 GND 1uF
C25 3V3_MN GND 22uF
R26 MN_PWR_EN 3V3 100k
TP15 3V3_MN 3V3_MN
C26 SW_PS_CT GND 1nF
C27 3V3_T35 GND 1uF
C28 PSRAM_VDD GND 4.7uF
C29 3V3 GND 100nF
R27 PSRAM_PWR_REQ GND 100k
R28 PSRAM_EN GND 100k
C30 PSRAM_SUP_CT GND 1nF
R29 PSRAM_SUP_MR_N 3V3_T35 100k
R30 PSRAM_PG 3V3_T35 10k
C31 3V3_T35 GND 100nF
TP16 PSRAM_VDD PSRAM_VDD
TP17 PSRAM_PG PSRAM_PG
TP18 PSRAM_EN PSRAM_EN
TP19 GND GND
TP20 GND GND
TP21 GND GND
TP22 GND GND
#FLG10 GND PWR_FLAG
FID1 fiducial
FID2 fiducial
FID3 fiducial
H1 GND M2.5
H2 GND M2.5
H3 GND M2.5
H4 GND M2.5

Generated by R08_WORK/cad/gen/gen_kicad.py from design.py — edit the source, not this file
SELECTED_UNVERIFIED engineering capture — NOT released for fabrication
STRUTHIO ARC—1
Sheet: /power/
File: power.kicad_sch
Title: ARC—1 Compact R08: Regulators, rail order, supervisors, staged—power
Size: User
Date: 2026—10—15
Rev: R08—draft
KiCad E.D.A. 9.0.9
Id: 3/10

F133-B SoC: supplies, clock, reset, pin assignment
Sheet generated from design.py; global labels carry every net.



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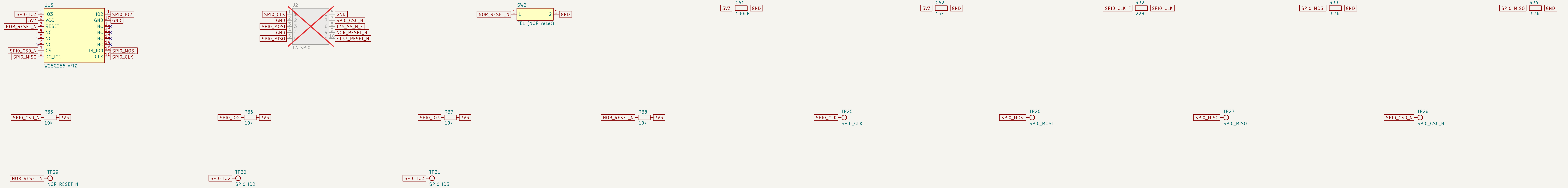
SELECTED_UNVERIFIED engineering capture — NOT released for fabrication
STRUTHIO ARC-1
Sheet: /f133/
File: f133.kicad_sch
Title: ARC-1 Compact R08: F133-B SoC: supplies, clock, reset, pin assignm
Size: User Date: 2026-10-15 **Rev: R08-draft**
KiCad E.D.A. 9.0.9 Id: 4/10

32 MiB SPI NOR boot (BOOT-SEL 00), straps, FEL/clip recovery
Sheet generated from design.py; global labels carry every net.



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STRUTHIO ARC-1

Sheet: /boot_nor/
File: boot_nor.kicad_sch

Title: ARC-1 Compact R08: 32 MiB SPI NOR boot (BOOT-SEL 00), straps, FE

Size: User
Date: 2026-10-15
Rev: R08-draft
KiCad E.D.A. 9.0.9
Id: 5/10

MN820 (BT820B + 128 MiB DDR3L) module socket, host SPI1, LVDS to R09 panel
Sheet generated from design.py; global labels carry every net.

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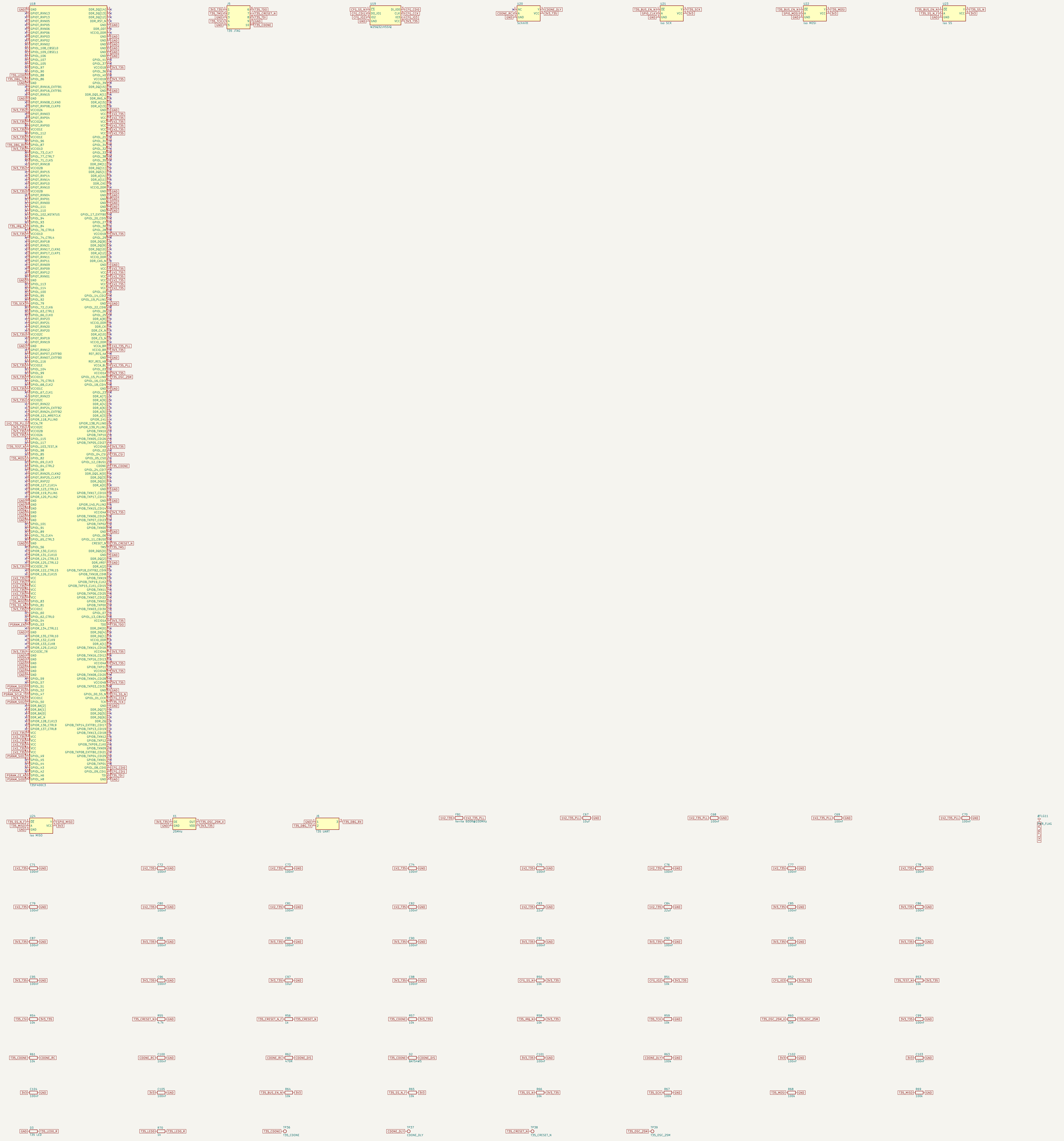
SELECTED_UNVERIFIED engineering capture — NOT released for fabrication
STRUTHIO ARC-1
Sheet: /mn820/
File: mn820.kicad_sch
Title: ARC-1 Compact R08: MN820 (BT820B + 128 MiB DDR3L) module sock
Size: User
Date: 2026-10-15
Rev: R08-draft
KiCad E.D.A. 9.0.9
Id: 6/10

T35F400C3 (optional accelerator): rails, config flash, SPI0 isolation, mailbox
Sheet generated from design.py; global labels carry every net.



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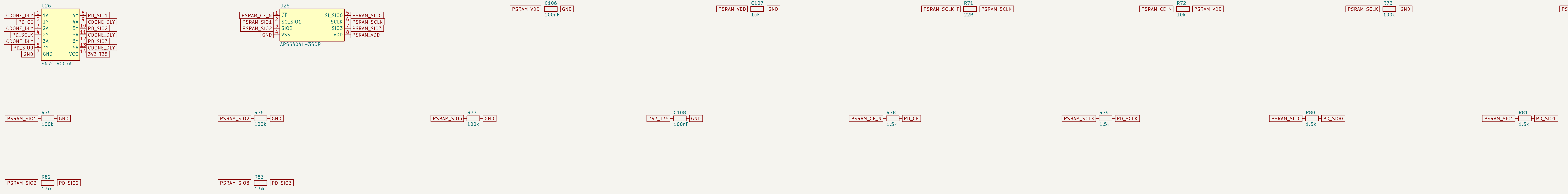
SELECTED_UNVERIFIED engineering capture — NOT released for fabrication
STRUTHIO ARC-1
Sheet: /t35/
File: t35.kicad_sch
Title: ARC-1 Compact R08: T35F400C3 (optional accelerator): rails, config f
Size: User
Date: 2026-10-15
Rev: R08-draft
KiCad E.D.A. 9.0.9
Id: 7/10

APS6404L 8 MiB PSRAM: switched VDD, switched start-up pull-downs
Sheet generated from design.py; global labels carry every net.



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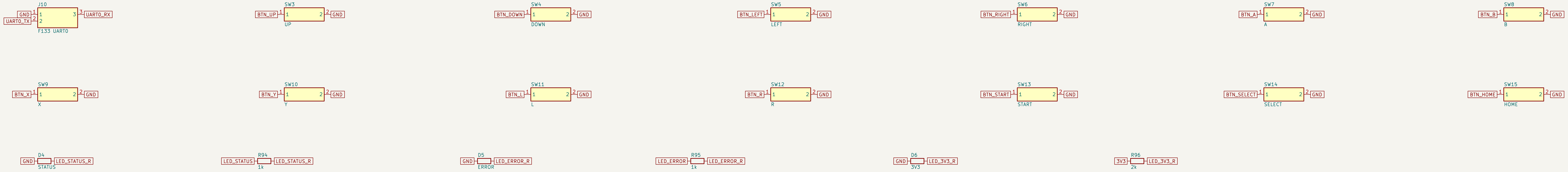
SELECTED_UNVERIFIED engineering capture — NOT released for fabrication
STRUTHIO ARC-1
Sheet: /psram/
File: psram.kicad_sch
Title: ARC-1 Compact R08: APS6404L 8 MiB PSRAM: switched VDD, switched
Size: User Date: 2026-10-15 **Rev: R08-draft**
KiCad E.D.A. 9.0.9 Id: 8/10

microSD game media (not boot), 2x MAX98357A I2S speakers
Sheet generated from design.py; global labels carry every net.

Generated by R08_WORK/cad/gen/gen_kicad.py from design.py — edit the source, not this file
SELECTED_UNVERIFIED engineering capture — NOT released for fabrication
STRUTHIO ARC-1
Sheet: /sd_audio/
File: sd_audio.kicad_sch
Title: ARC-1 Compact R08: microSD game media (not boot), 2x MAX98357A
Size: User
Date: 2026-10-15
Rev: R08-draft
KiCad E.D.A. 9.0.9
Id: 9/10

Game buttons, LEDs, debug UART
Sheet generated from design.py; global labels carry every net.



Generated by R08_WORK/cad/gen/gen_kicad.py from design.py — edit the source, not this file
SELECTED_UNVERIFIED engineering capture — NOT released for fabrication
STRUTHIO ARC-1
Sheet: /controls/
File: controls.kicad_sch
Title: ARC-1 Compact R08: Game buttons, LEDs, debug UART
Size: User
Date: 2026-10-15
Rev: R08-draft
KiCad E.D.A. 9.0.9
Id: 10/10